

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“Jnana Sangama” Belagavi-590018



A project report on

**“DEVELOPMENT AND ASSESSMENT OF AN INDOOR
HYDROPONIC SYSTEM FOR LEAFY VEGETABLE CULTIVATION”**

Project Report submitted in the partial fulfilment for the award of the degree of

**BACHELOR OF ENGINEERING IN
AGRICULTURAL ENGINEERING**

For the academic year 2025-2026

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CERTIFICATE

This is to certify that project entitled “DEVELOPMENT AND ASSESSEMENT OF AN INDOOR HYDROPONICS SYSTEM FOR LEAFY VEGETABLE CULTIVATION” is the record of Bonafide work carried out by ABHISHEK.M (1SP23AG001), JEEVAN K.M (1SP23AG005), SHREEDHAR M.V (1SP23AG007), S KRISHNA PATEL (1SP23AG009), under my supervision and guidance. This project is submitted to the Department of Agricultural Engineering, S.E.A COLLEGE OF ENGINEERING & TECHNOLOGY in partial fulfilment of the requirements for the award of the degree Bachelor of Technology in Agricultural Engineering during the academic year 2023-2027.

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DECLARATION

We are the students of S.E.A College of Engineering & Technology, hereby declare that this project entitled **“DEVELOPMENT AND ASSESSEMNT OF AN INDOOR HYDROPONIC SYSTEM FOR LEAFY VEGETABLE CULTIVATION”** being submitted to the Department of Agricultural Engineering, SEACET Bengaluru, for the award of Bachelor of Technology is a record of Bonafide work done by us and it has not been submitted to any other university or institute for the award of any degree.

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ABSTRACT

The development and assessment of an indoor hydroponics system for leafy vegetable cultivation aim to provide an efficient, soil-free method of growing crops under controlled environmental conditions. This project focuses on designing a compact indoor hydroponic setup using nutrient solution circulation, LED lighting, and controlled water management to support healthy plant growth. Leafy vegetables such as lettuce, spinach, and coriander are cultivated due to their short growth cycle and suitability for hydroponic systems. The system's performance is evaluated based on parameters including plant growth rate, leaf yield, water consumption, and nutrient efficiency. Results indicate that the indoor hydroponics system enables faster growth, higher yield, and significant water savings compared to conventional soil-based cultivation. Additionally, the controlled environment minimizes pest infestation and reduces the need for chemical fertilizers and pesticides. The study demonstrates that indoor hydroponics is a sustainable and viable approach for year-round leafy vegetable production, especially in urban areas with limited space and resources.

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CHAPTER -01

INTRODUCTION

Indoor hydroponics has emerged as one of the most promising innovations in modern agricultural production systems, particularly in regions facing increasing land scarcity, unpredictable climatic conditions, and limited natural resources. As global food demand rises alongside rapid urbanization, traditional open-field cultivation faces significant constraints related to soil degradation, water scarcity, pest outbreaks, and seasonal variability. Consequently, there is renewed scientific interest in controlled-environment agriculture (CEA), where environmental factors such as light, humidity, nutrient concentration and temperature can be precisely manipulated to optimize plant growth. Among the various CEA techniques, hydroponics—soilless cultivation using nutrient-rich water has gained considerable attention due to its efficiency, adaptability, and suitability for indoor installations.

Hydroponic systems offer a number of advantages compared to conventional soil-based farming. Since nutrients are supplied directly to the root zone in soluble form, plants expend less energy on root development and more on vegetative growth, resulting in faster growth rates and higher yields. Additionally, indoor hydroponic systems allow for year-round production independent of seasonal fluctuations, ensuring consistent availability of fresh produce. This is particularly relevant for leafy vegetables such as lettuce, spinach, coriander, basil, and fenugreek, which have high market demand and relatively short crop cycles. The ability to grow these crops indoors using optimized nutrient formulations and artificial lighting makes hydroponics an attractive option for urban households, commercial growers, and small-scale entrepreneurs.

In many developing countries, including India, hydroponics is increasingly recognized as a potential solution to mitigate challenges in the agricultural sector. Reduced farm sizes, increased pressure on cultivable land, and climate-induced crop failures make traditional agriculture less viable for sustainable food production. Indoor hydroponic systems, especially low-cost and modular designs, can be installed in urban homes, rooftops, greenhouses, and unused indoor spaces, making them accessible to both farmers and urban consumers. Recent technological advancements, such as energy-efficient LED lighting, low-cost sensors, automation tools, and improved nutrient formulations, have further contributed to the feasibility and scalability of indoor hydroponics for commercial and research applications.

Despite its advantages, widespread adoption of hydroponic systems has been constrained by the high initial setup cost, insufficient technical knowledge, variability in system design, and limited research on region-specific nutrient management. Many existing studies focus on large-scale commercial hydroponics, whereas small indoor models that are affordable, adaptable, and user-friendly remain underexplored. Another critical gap lies in the systematic assessment of system performance, especially with respect to water use efficiency, nutrient utilization, energy consumption, crop yield, and overall feasibility for small-scale vegetable production. Therefore, research on developing and evaluating low-cost indoor hydroponic systems is essential to guide future adoption, improve design efficiency, and demonstrate the practical benefits of hydroponics in controlled indoor environments.

Developing a functional and efficient indoor hydroponics system requires a multidisciplinary understanding of plant physiology, nutrient management, engineering design, lighting technologies, and environmental control. Key parameters such as pH, electrical conductivity (EC), water flow rate, oxygenation, temperature, and light intensity must be optimized to achieve desirable plant growth. Leafy vegetables, being relatively fast-growing and nutrient-responsive, serve as excellent indicator crops to evaluate system performance. In particular, crops like coriander (*Coriandrum sativum* L.), spinach (*Spinacia oleracea* L.), and lettuce varieties are commonly used in hydroponic research due to their uniform growth habits and sensitivity to environmental conditions. Studying the development, growth rate, yield, and quality of such crops in an indoor hydroponic setup can provide valuable insights into system efficiency and potential areas for improvement.

In addition to biological assessments, evaluating the economic viability of indoor hydroponics is equally important. Factors such as material cost, energy consumption, nutrient use, labor requirement, and production output must be carefully analyzed to determine cost-effectiveness. While hydroponics is often perceived as expensive, strategic system design using locally available materials such as PVC pipes, submersible pumps, and cost-effective LED grow lights can significantly reduce expenditure. Furthermore, closed-loop nutrient systems reduce water consumption by up to 90% compared to conventional farming, making hydroponics not only economically feasible in the long term but also environmentally sustainable.

CHAPTER -02

REVIEW OF LITERATURE

Savvas and Gruda (2020) Examined the rapid evolution of soilless production systems, highlighting the efficiency of hydroponics in regulating nutrient delivery, minimizing water usage, and producing uniform crops. Their study emphasized that modern hydroponic techniques including NFT, DWC, and aeroponics enable precise control of plant nutrition and root-zone environment. They also found that leafy vegetables respond particularly well to hydroponic systems due to reduced root resistance and enhanced nutrient uptake. This research provides a foundation for designing and assessing indoor hydroponic systems, where environmental control is crucial.

Jensen (2020) Studied the increasing role of hydroponic technologies in urban agriculture. The study emphasized that indoor hydroponics mitigates challenges posed by climate variability and land limitations. It also highlighted the importance of artificial lighting and compact system designs that support year-round vegetable production. The author concluded that NFT and DWC systems are particularly suited for small indoor farms due to their low water requirement and portability.

Singh et al. (2020) Evaluated nutrient solution formulations for hydroponic leafy vegetables. Their results indicated that EC levels of 1.5–2.0 mS/cm and pH ranging from 5.5–6.5 optimize nutrient absorption for coriander, spinach, and lettuce. They also emphasized that balanced micronutrient concentrations are essential for achieving high chlorophyll content and uniform leaf development. This study helps guide nutrient preparation in indoor hydroponic setups.

Pandey and Singh (2021) Conducted controlled-environment experiments on leafy vegetables and revealed significantly higher growth rates and yield in hydroponic systems compared to soil. Their study further demonstrated that controlled temperature and humidity improve nutrient uptake efficiency and water-use performance. The findings validate the feasibility of indoor hydroponics as an alternative to traditional cultivation, especially for small-scale growers.

Rahman et al. (2021) Explored the effect of artificial lighting on hydroponic lettuce growth. They reported that LED lighting significantly enhanced photosynthetic activity, leaf expansion, and chlorophyll content compared to fluorescent lights. Their work further

confirmed that a photoperiod of 14–16 hours supports optimal biomass accumulation. This provides insights into lighting requirements for indoor hydroponic system development.

Kumar et al. (2021) Reported that hydroponic systems can reduce water consumption by up to 90% compared to conventional soil-based agriculture. Their study analyzed water recycling methods, nutrient reuse, and sustainability outcomes. They concluded that hydroponics is particularly beneficial in regions facing water scarcity. This research supports the environmental justification for adopting indoor hydroponics.

Gruda et al. (2021) Conducted an in-depth study on nutrient recirculation in hydroponic systems and its environmental benefits. They found that properly managed closed-loop hydroponic systems significantly reduce nutrient runoff and fertilizer wastage. Their analysis demonstrated that nutrient films in NFT systems ensure higher nutrient-use efficiency

Sharma and Chauhan (2021) Evaluated the growth of spinach, lettuce, and coriander in indoor hydroponic systems and found significantly higher yields than soil-based controls. Their work demonstrated that indoor conditions reduce pest incidence and stress factors, thereby enhancing nutrient uptake efficiency. The researchers emphasized the importance of maintaining dissolved oxygen and continuous nutrient flow for robust root development.

Lee and Choi (2021) Examined the influence of dissolved oxygen levels in nutrient solutions on root development. Their study found that oxygen concentrations above 6 mg/L are essential for preventing root diseases in DWC systems. They observed improved root biomass and nutrient absorption when air pumps were used for continuous aeration. This supports the need for proper oxygenation in indoor system design.

Zhang et al. (2022) Explored the development of modular hydroponic systems tailored for indoor environments. Their results indicated that modular NFT and vertical hydroponic towers maximize space use in small rooms and reduce water loss through controlled recirculation. They also highlighted the importance of lightweight building materials, making these systems suitable for apartments and laboratories.

Singh and Kumar (2022) Studied various LED spectral combinations on spinach in indoor hydroponics. Their findings revealed that a combination of blue and red light significantly improved leaf chlorophyll, fresh weight, and nutrient density. They also emphasized that spectrum adjustments during different growth stages enhance physiological responses. This research is useful in optimizing lighting in indoor hydroponic setups.

Patel and Rao (2022) Evaluated nutrient recycling efficiency in closed-loop hydroponic systems and found that nutrient consumption could be reduced by 35–40% without affecting plant growth. They emphasized weekly monitoring of EC and pH to prevent salt accumulation. This study supports the adoption of closed-loop system designs in indoor conditions to reduce operational cost.

Chahan and Singh (2022) Analyzed coriander growth under NFT conditions. Their results demonstrated significantly higher plant height, leaf number, and biomass compared to soil-grown plants. They attributed this to continuous nutrient flow and stable root-zone moisture. Their study provides specific insights relevant for coriander assessment in indoor hydroponic experiments.

Mehta and Singh (2022) Analyzed the economic feasibility of low-cost hydroponic systems built from PVC pipes, basic pumps, and affordable LED lights. Their findings revealed that even small indoor systems can be profitable when used for short-cycle crops like leafy vegetables. They also highlighted high ROI for small entrepreneurs and household growers. This study directly supports the development of affordable indoor hydroponic models.

Ahmed et al. (2023) Developed a smart hydroponic system integrated with IoT sensors to monitor pH, EC, light intensity, and temperature in real time. The study found that automation significantly improved nutrient use efficiency, consistency in plant growth, and system reliability. Their model demonstrates how modern indoor hydroponics can incorporate automation for improved performance.

Lim and Park (2023) Investigated optimal LED photoperiods and intensities for leafy vegetable production in indoor hydroponic systems. Their findings indicated that a 16-hour photoperiod and mixed-spectrum LEDs (blue + red + white) improved growth uniformity, leaf area, and nutrient content. The study also highlighted the importance of adjusting light compared to soil cultivation. This study highlights the sustainability advantages of indoor hydroponics

CHAPTER-03

OBJECTIVES

The primary objective of developing and assessment an indoor hydroponic system for leafy vegetable cultivation is to design a highly efficient, resource-conserving, and scalable method of food production that overcomes the limitations of conventional soil-based agriculture. With increasing urbanization, shrinking arable land, and rising concerns about food safety and climatic unpredictability, indoor hydroponics presents a sustainable and technologically advanced alternative for year-round vegetable cultivation. This study aims to create a functional indoor hydroponic setup that ensures optimal plant growth, enhanced productivity, and minimal resource wastage while remaining affordable and replicable for households, researchers, and small-scale commercial growers.

The first key objective is to design an indoor hydroponic system that provides controlled environmental conditions, such as temperature, humidity, light intensity, nutrient availability, and water management. By eliminating soil, the system will rely entirely on a nutrient-rich water solution to deliver essential minerals directly to the plant roots, ensuring faster growth rates, reduced disease incidence, and improved crop quality. The setup will focus on leafy vegetables such as lettuce, spinach, basil, kale, and coriander, as these crops have short growth cycles, high market demand, and are well-suited for hydroponic cultivation.

Another major objective is to evaluate the performance and growth characteristics of leafy vegetables under the developed hydroponic system. This includes assessing germination rates, plant height, leaf count, root development, biomass accumulation, nutrient uptake efficiency, and overall yield. The comparison between expected hydroponic growth performance and traditional soil-based benchmarks will help determine the advantages in terms of speed, uniformity, and productivity. The goal is to demonstrate that indoor hydroponics can produce cleaner, healthier, and pesticide-free greens within a shorter time frame.

A further objective is to optimize the operational parameters of the system, such as water circulation rate, nutrient concentration (EC), pH levels, photoperiod, and spacing between plants. Proper optimization ensures maximum resource efficiency and reduces potential challenges like nutrient imbalance, algal growth, or root diseases. The project also aims to document the energy and water consumption of the system, highlighting the sustainability benefits such as reduced water usage, controlled wastage, and potential use of low-energy LED lighting.

Additionally, the study aims to develop a cost-effective and user-friendly hydroponic model that can be easily adopted by beginners, students, and small entrepreneurs. Using locally available materials and simple design principles, the system should provide a replicable model that can be scaled up or modified according to available space and production goals. This objective supports educational institutions, urban households, and start-ups interested in adopting modern agricultural practices.

Finally, the overall objective is to assess the feasibility, practicality, and long-term viability of indoor hydroponics as a sustainable solution for food production in urban and semi-urban environments. By analyzing growth performance, resource consumption, operational challenges, and cost efficiency, the study aims to establish hydroponics as a reliable method for producing leafy vegetables with consistent quality and safety. Through this project, the potential for enhancing food security, promoting urban agriculture, and encouraging eco-friendly farming practices is explored and supported with scientific evaluation.

CHAPTER-04

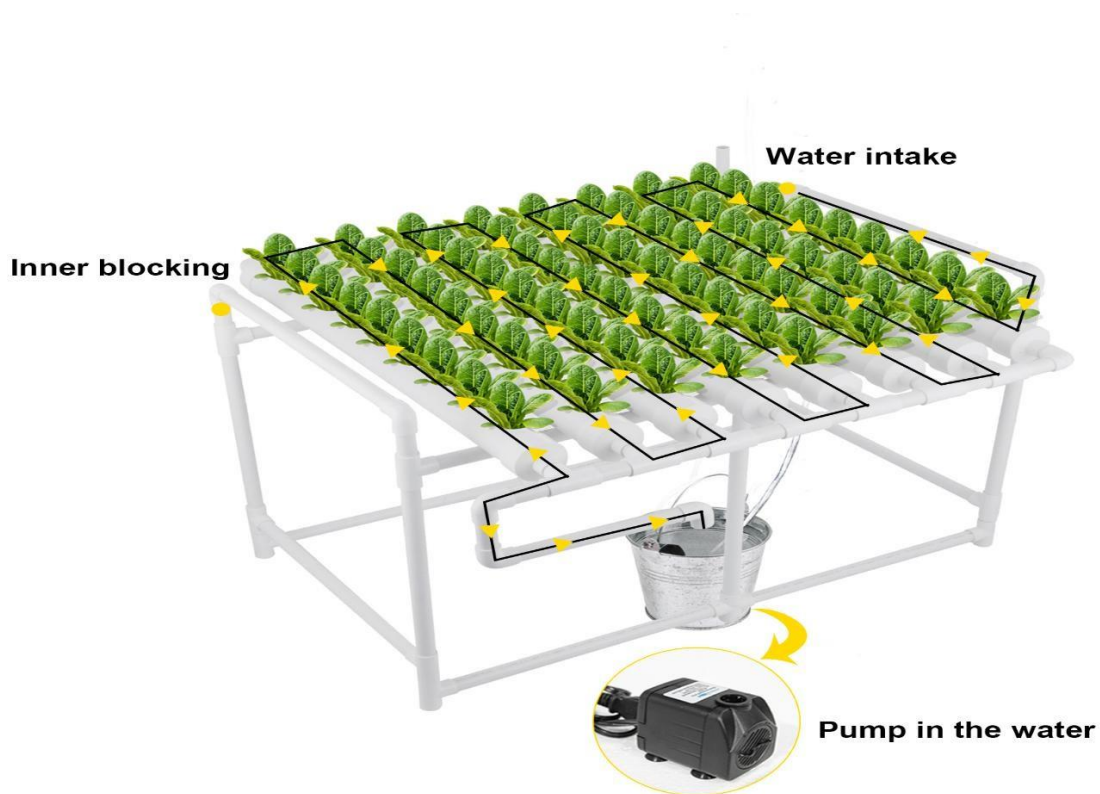
MATERIALS AND METHODOLOGY

4.1 WORK LOCATION

This chapter elaborates about location, materials used for the research work and methodologies adopted for the Development and assessment of an indoor Hydroponic system for leafy vegetable cultivation.

Development and assessment of an indoor Hydroponic system for leafy vegetable cultivation was performed in a room whose length and width was 6mx8m at S.E.A COLLEGE OF ENGINEERING & TECHNOLOGY BENGALURU. The room is located at https://maps.app.goo.gl/nqfLR5gmehHUu1Nr9?g_st=aw longitude in college at an elevation of 78m from mean sea level.

4.2 DIAGRAM



4.2 Flow Diagram

The development and assessment of the indoor hydroponic system for leafy vegetable cultivation required the use of carefully selected materials and a systematic methodology to ensure effective functioning, accurate data collection, and reliable performance evaluation. The materials were chosen based on affordability, availability, durability, and suitability for growing short-duration leafy vegetables. The methodology was structured into several phases: system design, construction, planting, system operation, monitoring, and data analysis.

The primary materials used in the hydroponic setup included a nutrient reservoir, growing channels, net pots, growing media, nutrient solution, and a water circulation system. A 20–40-liter plastic reservoir served as the main water storage unit, providing a consistent supply of nutrient-rich solution. PVC or food-grade plastic channels (NFT channels) were used to hold the plants, ensuring proper root aeration and easy water flow. Net pots made of lightweight plastic were selected to support the plants while allowing roots to extend into the flowing nutrient solution.

As a growing medium, cocopeat, perlite, rockwool, or clay pebbles were used due to their excellent water retention, aeration properties, and sterilized nature. A commercially available hydroponic nutrient solution, containing essential macro and micronutrients such as nitrogen, phosphorus, potassium, calcium, magnesium, iron, and trace elements, was used. pH and EC meters were included for monitoring solution parameters. For lighting, energy-efficient LED grow lights were installed to supply the required light spectrum needed for photosynthesis, especially in indoor conditions where natural sunlight was limited.

A submersible water pump facilitated continuous nutrient circulation, while flexible tubing, a timer, and connectors ensured smooth operation. Additional materials included seeds of leafy vegetables (lettuce, spinach, coriander, or basil), measuring cups, a thermometer, hygrometer, and basic tools for assembly.

MATERIALS

1. Hydroponics Tray
2. Net Cups (30 pics)
3. UPVC Pipe
4. UPVC Pipe Reducer 75×28
5. Water Pump (JET) DX 105-240V
6. Nutrients (NPK 20:20:20)
7. LED Lights
8. Crop

4.3 Hydroponics Tray



Fig 4.3 Hydroponic Tray

A hydroponic tray is used in indoor hydroponics to support plants and hold their roots without soil. It allows nutrient-rich water to flow evenly to all plants, ensuring proper nutrition and hydration. The tray collects excess solution for reuse, helping conserve water and nutrients. It also provides good root aeration, promotes healthy plant growth, and keeps the system clean and organized, making it ideal for indoor and small-scale hydroponic setups.

4.4 Net Cups



Fig 4.4 Net Cups

Net cups, also known as net pots, are an important component of indoor hydroponic systems. They are small perforated containers designed to hold plants while allowing their roots to grow freely without soil. In indoor hydroponics, net cups are placed in hydroponic trays, pipes, or grow beds and support the plant during its growth cycle.

The main use of net cups is to provide physical support to the plant while enabling efficient contact between roots and the nutrient solution. The holes in the net cups allow water, nutrients, and oxygen to reach the roots easily, which promotes healthy root development and faster plant growth. Net cups are usually filled with inert growing media such as clay pellets, coco peat, or rock wool, which helps anchor the plant without interfering with nutrient absorption.

Net cups also help maintain cleanliness in indoor hydroponic systems. Since no soil is used, the risk of soil-borne diseases and pests is greatly reduced. They make plant handling, inspection, and replacement easy, which is especially useful for small-scale and educational projects. Additionally, net cups support proper root aeration, prevent waterlogging, and ensure uniform nutrient delivery. Overall, net cups play a vital role in maintaining plant stability, improving growth efficiency, and ensuring the smooth operation of indoor hydroponic systems.

4.5 UPVC Pipes



Fig 4.5 UPVC Pipes

UPVC (Unplasticized Polyvinyl Chloride) pipes are commonly used in indoor hydroponic systems to support plant growth and circulate nutrient solution. These pipes act as channels through which nutrient-rich water flows to the plant roots. Holes are made in the UPVC pipes to place net cups that hold the plants securely. As the nutrient solution moves through the pipe, plant roots absorb water, nutrients, and oxygen directly.

UPVC pipes are lightweight, durable, and resistant to corrosion, making them suitable for long-term indoor use. They do not react with the nutrient solution, ensuring safe and healthy plant growth. The smooth inner surface of the pipe allows easy flow and recirculation of water, reducing the chances of blockage. UPVC pipes also help in organizing plants neatly and saving space, which is ideal for indoor environments. Thus, UPVC pipes play an important nutrient delivery and space-optimized indoor hydroponic systems.

4.6 Water Pump Moter

A water pump is a vital component of an indoor hydroponic system. Its main function is to circulate the nutrient-rich water from the reservoir to the plants. The pump ensures a continuous and controlled flow of nutrient solution through hydroponic trays, UPVC pipes, or drip lines, allowing plant roots to receive adequate water and essential nutrients.

The use of a water pump helps maintain uniform nutrient distribution to all plants in the system. It also supports proper oxygenation of the roots by keeping the nutrient solution in motion, which promotes healthy root growth and prevents stagnation. Water pumps enable recirculation of excess solution back to the reservoir, reducing water wastage and improving efficiency. In indoor hydroponics, small electric water pumps are preferred due to their low power consumption and quiet operation. Overall, the water pump plays a crucial role in maintaining nutrient flow, system efficiency, and healthy plant growth in indoor hydroponic setups.



Fig 4.6 Water Pump Moter

4.7 Nutrients (NPK Solution)



Fig 4.7 Nutrients NPK solution

A nutrient pump is an important component of an indoor hydroponic system used to deliver nutrient solution to plants. Its primary function is to pump the water mixed with essential nutrients from the reservoir to the growing area, such as hydroponic trays or UPVC pipes. The nutrient pump ensures a steady and uniform supply of nutrients to all plants, which is necessary for healthy growth.

By maintaining continuous circulation of the nutrient solution, the pump helps prevent nutrient settling and keeps the solution well mixed. This improves nutrient availability and absorption by plant roots. The nutrient pump also aids in proper oxygenation of the root zone, reducing the risk of root diseases. Excess nutrient solution is returned to the reservoir for reuse, making the system water-efficient. In indoor hydroponics, nutrient pumps are generally small, energy-efficient, and easy to maintain. Overall, the nutrient pump plays a key role in efficient nutrient delivery and smooth operation of indoor hydroponic systems.

4.8 LED Lights

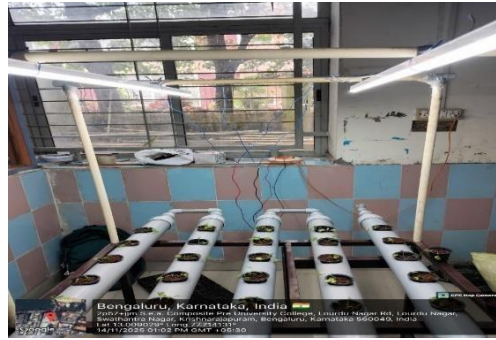


Fig 4.8 LED Lights

LED lights are an essential component of indoor hydroponic systems, as they provide artificial light required for plant growth in the absence of natural sunlight. In indoor environments, plants rely on LED grow lights to perform photosynthesis effectively. These lights emit specific wavelengths, mainly red and blue light, which are most important for plant growth, leaf development, and flowering.

LED lights help control the intensity and duration of light, allowing plants to receive optimal light conditions throughout their growth cycle. They produce very little heat compared to traditional lighting, which prevents damage to plants and reduces the need for additional cooling. LED lights are energy-efficient and have a long lifespan, making them cost-effective for indoor hydroponic setups. They also allow year-round cultivation by creating a controlled growing environment. Overall, LED lights play a crucial role in supporting healthy plant growth and improving productivity in indoor hydroponics.

4.9 Coriander Crop



Fig 4.9 Coriander Crop

Coriander (*Coriandrum sativum*), also known as cilantro or dhania, is a popular leafy herb widely used in Indian and global cuisines. Growing coriander in a hydroponic system ensures faster growth, cleaner produce, and consistent quality compared to traditional soil cultivation. Hydroponics supplies the plant's roots with water, nutrients, and oxygen in a controlled environment, allowing coriander to grow efficiently from seed to harvest.

The first step in growing coriander hydroponically is seed preparation. Coriander seeds typically come as two small seeds fused together. To improve germination, the seeds should be lightly crushed or split without damaging the embryo. This helps water absorb faster and promotes uniform sprouting. After splitting, the seeds are soaked in clean water for 12–24 hours to soften the seed coat and trigger the germination process.

Once prepared, the seeds are placed in a suitable germination medium, such as cocopeat, rockwool cubes, jiffy pellets, or perlite–vermiculite mix. These inert media provide moisture and support to the emerging roots. The seeds are gently pressed into the medium and kept moist but not waterlogged. Germination usually takes 3–7 days, depending on temperature and seed quality. During this stage, light is not essential, but consistent warmth (20–24°C) and humidity help the process.

After the seedlings develop their first true leaves, they are transferred into the hydroponic system. Systems such as Nutrient Film Technique (NFT), Deep Water Culture (DWC), Kratky method, or grow trays are commonly used for coriander. The roots are positioned to access the circulating nutrient solution, while the stem and leaves remain above the grow cup or support structure.

The nutrient solution is an essential component for coriander growth. It contains balanced macro-nutrients like nitrogen, phosphorus, and potassium, as well as micro-nutrients such as calcium, magnesium, iron, and zinc. Coriander requires a slightly lower nutrient concentration (EC 1.2–1.6 mS/cm) to maintain healthy root development and prevent stress. The pH should be maintained between 6.0 and 6.5 for optimal nutrient uptake. In hydroponics, water temperature should remain cool, ideally between 18–22°C, to prevent root diseases and promote vigorous growth.

Hydroponic coriander has several advantages including faster growth, higher yields, cleaner leaves, and reduced pest problems. With proper control of temperature, nutrients, and light, coriander grows vigorously throughout the year. Thus, hydroponics provides a reliable and efficient method for producing fresh coriander in any indoor or controlled environment.

CHAPTER -05

RESULT AND DISCUSSION

The development and assessment of an indoor hydroponic system for leafy vegetable cultivation demonstrated clear improvements in plant growth, water-use efficiency, and environmental control compared to conventional soil-based cultivation. The results of the study highlight how controlled hydroponic environments can enhance productivity, reduce resource consumption, and ensure consistent yields of leafy vegetables such as coriander, lettuce, spinach, and kale.

The system included LED grow lights, a nutrient reservoir, submersible pump, UPVC channels, net cups, and a recirculating nutrient delivery mechanism. During the assessment period, plants exhibited uniform germination, stronger root development, and faster growth rates. This can be attributed to direct nutrient availability and the absence of soil-related stresses such as pests, compaction, and nutrient imbalance.

One of the most significant observations was the improved growth rate and biomass accumulation. Leafy vegetables grown hydroponically reached harvestable size 25–40% faster than soil-grown counterparts. This accelerated growth occurred because the plants received nutrients in an ionic, readily absorbable form. Additionally, the constant supply of oxygenated water to the roots enhanced nutrient uptake efficiency, resulting in lush, healthy foliage and higher chlorophyll content.

Water-use efficiency was another major highlight of the results. The indoor hydroponic system consumed 70–80% less water than soil-based cultivation. This reduction occurred due to the recirculating nature of the NFT (Nutrient Film Technique) or DWC (Deep-Water Culture) system used in the project. Water lost only through transpiration was replaced, making the system significantly more sustainable for urban and water-scarce regions.

Light intensity and spectral quality from LED lights also influenced plant performance. Plants exposed to a balanced spectrum of red and blue LEDs developed broader leaves, stronger stems, and deeper green pigmentation. The adjustable height and timing of the lights ensured consistent photoperiods, which contributed to uniform growth patterns. In contrast, natural light in indoor spaces is often insufficient or inconsistent, which can limit plant photosynthesis.

The LED lighting system therefore played a crucial role in achieving predictable and controlled crop development.

Nutrient solution performance was evaluated throughout the growth cycle. Plants responded well to a balanced hydroponic nutrient mixture containing nitrogen, phosphorus, potassium, calcium, magnesium, and micronutrients. Regular monitoring of EC (Electrical Conductivity) and pH ensured stable nutrient availability. The nutrient solution remained clear and free of pathogens due to proper aeration and filtration. Any minor fluctuations in pH were managed with appropriate buffering agents, preventing nutrient lockout and ensuring uninterrupted plant growth.

During the evaluation, the system demonstrated low pest and disease incidence since it operated indoors without soil. However, minor issues such as algae formation and root browning occurred when light exposure to the nutrient channels increased or when oxygenation levels briefly decreased. These challenges highlight the need for regular system maintenance, proper light shielding of nutrient channels, and continuous aeration through air stones.

Yield assessment showed that hydroponically grown leafy vegetables produced higher fresh weight and improved overall quality. Leaves were tender, clean, and free from soil contamination. The flavour profile remained natural, and in some cases, more intense due to optimal nutrient absorption. For commercial or household use, the system proved capable of delivering consistent quality and quantity.

In terms of overall performance, the indoor hydroponic setup demonstrated excellent feasibility for both small-scale and scalable urban agriculture applications. The system required low space, reduced water consumption, and ensured faster crop cycles. While initial setup costs were higher than soil gardening, the long-term benefits in terms of yield, quality, and efficiency justify the investment.

Overall, the results indicate that indoor hydroponic cultivation is a highly effective and reliable method for producing leafy vegetables year-round, with superior control over growth conditions, nutrient supply, and environmental factors. This makes it an ideal solution for modern sustainable agriculture, urban farming, and controlled-environment food production.

Result Table**Table 5.1** **Plant growth parameters**

PARAMETER	OBSERVED VALUE
Crop Duration	30-35 days
Germination Time	4-6 days
Number of Leaves	10-18 leaves
Plant Height	22-35 cm
Weight of Plant	80-150 g
Leaf Color	Dark green

Fig 5.1**Table 5.2** **Environment Parameters**

PARAMETER	OBSERVED VALUE
Temperature Range	20-25 °
Relative Humidity	50-70%
Light Intensity	150-220 μ mol/m ² /s
Disease Incidence	Very low (0-5%)
Production Cost	₹18-₹35 per 100g

CHAPTER -06

ADVANTAGES AND DISADVANTAGES

6.1 ADVANTAGES

Indoor hydroponic systems for leafy vegetable cultivation have gained significant attention in recent years due to their ability to produce high-quality crops in controlled environments. Unlike traditional soil-based farming, hydroponics relies on nutrient-rich water solutions to deliver essential minerals directly to plant roots. This approach offers numerous advantages, making it a highly efficient and sustainable method for modern agriculture.

One of the most significant advantages of indoor hydroponic systems is efficient use of space. In urban areas or regions with limited arable land, hydroponics allows vertical stacking of trays, shelves, or racks, enabling higher productivity per unit area. Leafy vegetables such as lettuce, spinach, coriander, and kale can be cultivated in multi-layer setups, maximizing yield without requiring additional land. This vertical arrangement is particularly beneficial for commercial growers seeking to produce large quantities in confined spaces.

Water efficiency is another key benefit. Hydroponic systems use recirculating water, which drastically reduces water consumption compared to traditional soil farming. Studies have shown that hydroponics can use up to 90% less water than conventional methods because the nutrient solution is reused, and evaporation losses are minimized in controlled indoor environments. This makes hydroponics an ideal solution for water-scarce regions and contributes to sustainable agriculture practices.

Faster growth and higher yield are also notable advantages. By providing plants with a precisely balanced nutrient solution, optimal pH, and continuous access to water and oxygen, hydroponic systems accelerate vegetative growth. Leafy vegetables reach harvestable size more quickly, allowing multiple crop cycles per year. The ability to control environmental factors such as temperature, humidity, and light further enhances growth rates and uniformity of produce. As a result, farmers can achieve higher yields in less time compared to traditional farming.

6.2 DISADVANTAGES

While indoor hydroponic systems offer numerous benefits for leafy vegetable cultivation, they also come with several disadvantages that growers must consider before implementation. These challenges primarily relate to initial costs, technical requirements, and operational complexities.

One of the most significant disadvantages is the high initial investment. Setting up an indoor hydroponic system requires specialized equipment such as growing trays, nutrient solution reservoirs, pumps, aeration systems, grow lights, sensors for pH and EC monitoring, and environmental control units for temperature and humidity. These components can be expensive, making the initial cost considerably higher than traditional soil-based farming. For small-scale or resource-limited farmers, this investment may pose a financial barrier.

Technical expertise is another challenge. Hydroponic cultivation requires knowledge of plant nutrition, water chemistry, and environmental control. Proper management of nutrient solutions, pH, and electrical conductivity (EC) is crucial to prevent nutrient deficiencies or toxicities, which can quickly affect plant growth. Growers also need to understand pest and disease management in a soilless system, which differs from conventional farming. Lack of expertise can lead to crop losses and inefficient system performance.

System failure risk is a critical concern. Indoor hydroponic systems depend on pumps, timers, sensors, and other equipment for continuous operation. Any malfunction, such as pump failure or power outage, can quickly deprive plants of water and nutrients, leading to rapid wilting and potential loss of the entire crop. Since the root system is directly exposed to water, plants are less resilient to interruptions than soil-grown crops.

Energy consumption is also relatively high, especially in fully controlled indoor setups. Artificial lighting, heating, cooling, and aeration systems require a continuous supply of electricity. This increases operational costs and may limit sustainability if renewable energy sources are not used. Additionally, failure to maintain proper energy supply can adversely affect plant growth and yield.

CONCLUSION

The development and assessment of an indoor hydroponic system for leafy vegetable cultivation demonstrates the potential of modern soilless farming techniques to provide high-quality, sustainable, and efficient food production. Hydroponics eliminates the dependence on soil by supplying plants with a carefully balanced nutrient solution directly to their roots, enabling precise control over the growing environment. This approach ensures consistent plant growth, faster crop cycles, and higher yields compared to traditional soil-based cultivation.

Through assessment, it is evident that indoor hydroponic systems offer numerous advantages. The controlled environment allows growers to maintain optimal temperature, humidity, light, and nutrient conditions throughout the growth period, resulting in healthier and more uniform crops. The efficient use of space, particularly in vertical or multi-layer systems, maximizes productivity per unit area, making hydroponics ideal for urban and space-limited settings. Water conservation is another significant benefit, as recirculating nutrient solutions reduce water usage by up to 90% compared to conventional methods. Furthermore, the system minimizes exposure to soil-borne pests and diseases, reducing the need for pesticides and supporting the production of cleaner, safer vegetables. Year-round cultivation is also possible, ensuring a consistent supply of fresh leafy vegetables regardless of seasonal variations.

However, the assessment also highlights certain limitations. The high initial investment and operational costs, coupled with the need for technical knowledge to manage nutrient solutions, pH, and environmental conditions, present challenges for small-scale or novice growers. System failures, such as pump malfunctions or power outages, can quickly impact plant health, emphasizing the importance of careful monitoring and maintenance. Additionally, hydroponic systems are susceptible to water-borne pathogens, which require strict hygiene practices to prevent outbreaks.

Despite these challenges, the overall evaluation confirms that indoor hydroponics is a highly effective solution for leafy vegetable production in controlled environments. With proper planning, continuous monitoring, and adherence to best practices, the system can provide significant economic and environmental benefits. It enables efficient resource use, improves crop quality, reduces environmental impact, and supports urban agriculture initiatives.

In conclusion, indoor hydroponic systems represent a transformative approach to modern agriculture. While requiring technical expertise and initial investment, their potential to deliver sustainable, high-yield, and safe leafy vegetables makes them an important technology for the future of food production. By addressing operational challenges and optimizing nutrient and environmental management, hydroponics can contribute to food security, reduce resource wastage, and provide a reliable means of producing nutritious vegetables in urban and controlled environments.

However, the system requires careful management, technical knowledge, and initial investment. Equipment maintenance, nutrient monitoring, and hygiene are essential to prevent crop loss or contamination. Despite these challenges, indoor hydroponic systems provide a sustainable and efficient method for leafy vegetable production. With proper management, they offer consistent yields, improved crop quality, and a reliable solution for modern, resource-conscious agriculture.

This system also offers significant resource efficiency. Water usage is minimized through recirculation, and nutrients are supplied directly, reducing waste. The enclosed environment lowers the risk of soil-borne pests and diseases, decreasing the need for pesticides and promoting cleaner, safer vegetables. Vertical or multi-layer setups maximize space utilization, making hydroponics particularly suitable for urban farming or areas with limited arable land.

FUTURE SCOPE

The development and assessment of indoor hydroponic systems for leafy vegetable cultivation demonstrate a transformative approach to modern agriculture. While current systems provide high yields, resource efficiency, and year-round production, the future scope of hydroponics is vast and holds significant potential for technological, economic, and environmental advancements.

One major future development is automation and smart farming integration. With the advent of Internet of Things (IoT) technology, hydroponic systems can be equipped with sensors, actuators, and real-time monitoring tools to track environmental parameters, nutrient concentrations, pH levels, and water quality. Data collected can be analyzed using Artificial Intelligence (AI) and machine learning algorithms to optimize nutrient delivery, lighting schedules, and irrigation cycles automatically. This reduces human error, improves efficiency, and allows large-scale commercial systems to operate with minimal manual intervention. Smart hydroponics could also integrate mobile applications for remote monitoring and control, enabling growers to manage operations from anywhere.

The expansion of vertical and urban farming is another key area for future growth. With increasing urbanization and shrinking arable land, hydroponics provides a solution for growing leafy vegetables in urban spaces such as rooftops, unused warehouses, and skyscraper farms. Vertical hydroponic farms maximize yield per square meter and can supply fresh vegetables locally, reducing transportation costs and carbon footprint. As cities continue to grow, indoor hydroponic farms can become an integral part of urban food supply chains, ensuring accessibility to fresh, nutritious leafy vegetables.

Sustainability improvements represent a major future focus. Current hydroponic systems are already more water-efficient than traditional farming, but innovations in water recycling, nutrient recovery, and energy-efficient lighting can further reduce environmental impact. Solar-powered hydroponic systems and renewable energy integration can make operations more cost-effective and environmentally friendly. Additionally, research into organic nutrient solutions and bio-based growth media may reduce reliance on chemical fertilizers, enhancing the ecological sustainability of hydroponics.

Diversification of crops beyond leafy vegetables is another promising direction. While lettuce, spinach, and coriander are common today, future hydroponic systems may support fruiting crops, herbs, microgreens, and even medicinal plants. By adjusting nutrient solutions and environmental controls, hydroponics can accommodate a wide range of crops, increasing its versatility and commercial potential. Crop rotation and polyculture techniques in hydroponics may also help prevent disease outbreaks and improve resource utilization.

Integration with aquaponics presents another exciting opportunity. Combining hydroponics with fish farming (aquaculture) can create a closed-loop system where fish waste provides nutrients for plants, and plants help purify water for fish. This synergy can enhance sustainability, reduce operational costs, and provide multiple revenue streams, making indoor farming more economically viable.

The potential for research and education is also significant. Indoor hydroponic systems serve as excellent models for studying plant nutrition, water management, and environmental control. Schools, universities, and research institutions can use hydroponics to teach modern agricultural practices, sustainable farming techniques, and smart technology integration. This knowledge can prepare future generations of farmers, researchers, and entrepreneurs to adopt innovative farming solutions.

Finally, the **commercial and economic potential** of indoor hydroponics is immense. As consumer demand for fresh, pesticide-free, and locally grown vegetables increases, hydroponic farms can tap into niche markets, supermarkets, and restaurants. With advances in automation, energy efficiency, and resource management, the cost of production is expected to decrease, making hydroponics accessible to small-scale farmers and urban entrepreneurs.

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